



Improvement of Rainfall Measurements by Using a Dual Tipping Bucket Rain Gauge

Jeong Hwan Choi¹ · Ki-Ho Chang¹ · Kyung-Eak Kim¹ · Ki Seok Bang²

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Abstract

The Dual Tipping Bucket Gauge (DTBG) is newly developed to improve the accuracy of rainfall measurements. DTBG includes two tipping buckets (TBs) in a gauge cylinder, named TB01 and TB05. The measurement resolution of TB01 and TB05 are 0.1 mm and 0.5 mm, respectively. Rainfall measurements by DTBG are made simultaneously by the two TBs. The higher amount of rainfall from either TB01 or TB05 is then taken as the rainfall amount by DTBG, which constitutes a major advantage of DTBG compared to a single TB rain gauge. For 14 rainfall events, the accuracy of DTBG was assessed by inter-comparisons of rainfall amounts by DTBG and Pluvio2 (reference gauge). The rainfall intensities by DTBG were fairly consistent with those by Pluvio2, with an average fractional bias of 0.07%. The present study demonstrates that DTBG is more accurate and reliable compared to a single TB rain gauge.

Keywords Dual tipping bucket rain gauge · Improvement of rainfall measurements · Inter-comparison

1 Introduction

Rainfall measurement by rain gauges is critical to elucidate changes of water budget, water resources, and climate, including hazards due to heavy rainfall (Vörösmarty et al. 2000; Frich et al. 2002; Oki and Kanae 2006; Trenberth 2011). Rain intensity data at the ground are essential to remote sensing techniques because the data are usually used as the “ground truth” in rainfall calibration for remote sensing by meteorological satellite and radar (Brandes 1975; Zawadzki 1975; Collier 1986; Austin 1987; Ciach et al. 1997; Bowman et al. 2003; Kneis et al. 2014;). The data are also crucial to the validation of mesoscale quantitative precipitation forecasts (Yates et al. 2006). Extensive applications require high accuracy and temporal resolution of the data. However, it is challenging to make accurate measurements of rainfall intensity because rainfall possesses an

intermittent nature, as well as high spatial and temporal variability, including sampling properties of rain gauges and measurement errors (Ciach and Krajewski 1999).

Measurements of rainfall at the ground level are normally made by tipping bucket (TB) rain gauges, especially for automatic observations in national weather and rainfall stations (Marsalek 1981; WMO 2008). The reason for this is that they can be easily maintained and are durable. In addition, TB rain gauges are relatively inexpensive compared to other types of rain gauges, such as weighing gauges and optical rain gauges. Rainfall measurements by TB rain gauges, however, are frequently subject to a variety of errors (WMO 2008).

The errors are usually divided into two types: random and systematic errors. The random errors are experimental uncertainties that can be revealed by repeating measurements; those that cannot be revealed in this way are called systematic (Taylor 1982). The random errors can be estimated by repetition of the experiment or observation on a given subject. Based on the definition of two errors, all errors related rainfall measurements by rain gauges are considered to be systematic since every rainfall has its own characteristics under a large variety of occurrence condition. There are many kinds of systematic errors as the followings. Some of the errors comprise rainfall losses attributable to wind, evaporation, wetting, and splash-out. There are also

Responsible Editor: Kei Yoshimura

✉ Ki-Ho Chang
khchang@korea.kr

¹ National Institute of Meteorological Sciences/KMA,
Jeju 63568, South Korea

² GBM Inc., Urime-biz center 1st 1413, Seoul 08377,
South Korea

systematic errors due to malfunctions of TBs (no tipping or improper tipping of buckets), calibration effects, and sitting and exposure of rain gauges (Habib et al. 2010). Errors by TB rain gauges are mainly ascribed to the undercatchment of rainfall, of which there are several causes. During rainfall, TB cannot reposition itself fast enough after a tip to collect all of the rain water from the funnel above the TB.

As a consequence, rainfall is not measured during the finite time required for a bucket to tip from one side to the other (Duchon and Essenberg 2001; Duchon and Biddle 2010). TB rain gauges also undercatch under high rain rates as water continues to pour into the already filled bucket during the bucket motion, causing overflow of water from the bucket. Indeed, undercatchment has been reported to be significant for rain intensity greater than 25 mm/h (Humphrey et al. 1997; Marsalek 1981; WMO 2008). Furthermore, undercatchment increases nonlinearly with rain intensity (Humphrey et al. 1997). Nystuen (1999) examined the performance of six different types of automatic rain gauges, including TB, weighing, capacitance, and optical and acoustical sensors, under various rainfall conditions. It was found that TB rain gauges exhibit a tendency to underestimate rainfall amount during extremely high rainfall intensity (over 100 mm/h), and show significant instrumental noise for light rainfall intensity (under 2 mm/h). Duchon and Biddle (2010) demonstrated that when rain intensities exceed approximately 50 mm/h, TB rain gauges noticeably undercatch rainfall totals. In addition, TB rain gauges possess a problem in estimation of rainfall amounts in the following cases. When the first period of rain stops prior to tipping the bucket with some amount of water, TB gauges underestimate the rainfall amount as less than the actual amount. Then, when the next period of rain starts, the bucket tips by taking in a small amount of water. This makes it possible to overestimate the rainfall compared to the actual amount. This kind of error will be more frequent when the volume of the bucket is large (Das and Prakash 2011).

Undercatchment also occurs due to wind, wetting loss on the internal walls of the gauge, and splash-out, including evaporation from the container and deficiency in gauge calibration. According to Nespor and Sevruk (1999), wind-induced rainfall loss can be 2%–10% for rain and 10%–50% for snow, on average. Duchon and Biddle (2010) showed that the effect of wind on rainfall undercatchment is significant only when its speed at a height of 2 m is greater than approximately 5 ms^{-1} – 6 ms^{-1} . To reduce undercatchment, wind shields are commonly employed. Undercatchment due to rainfall loss by wetting and evaporation is also a problem that is common to all gauges. Undercatchment attributable to wetting is typically around 0.05 mm per rainfall event (Niemczynowicz 1986). Undercatchment due to evaporation may be on the order of 0.004 mm/h (Fankhauser 1998). The splash-out of rain drops from the surface of open

collector (refer to Fig. 3) which is not collected also leads to undercatchment of rainfall. The error ascribed to splash-out is approximately 1%–2% (Lanza et al. 2005; WMO 2008; Rodda and Dixon 2012). These errors are commonly accounted for by means of correction models (Sevruk 1982; Legates and Willmott 1990).

TB rain gauges also often suffer from unpredictable gauge problems, such as mechanical or electrical issues. In this case, normal data acquisition cannot be expected during the period of rainfall. To detect the gauge problem, as well as to secure normal data acquisition, one pair of rain gauges with the same bucket size was recommended (Krajewski et al. 1998; Ciach and Krajewski 1999). In addition, the accuracy of rainfall measurements can be enhanced by using two TB rain gauges with different bucket sizes. This approach offers the major advantage of compensating for their shortcomings, and promotes more accurate measurements of rain intensity, rainfall amount, and rainfall duration. Since 1998, the KMA (Korea Meteorological Agency) has used two TB rain gauges at its weather stations: TBRG01 and TBRG05. TBRG01 and TBRG05 here represent TB rain gauges whose sizes are 0.1 mm and 0.5 mm, respectively. TBRG01 has been used for the measurement of rainfall amounts less than 0.5 mm and rainfall duration. This is because TBRG01 has a higher measurement resolution than TBRG05. TBRG05, on the other hand, has been used only for the measurement of rainfall amounts greater than or equal to 0.5 mm. This is because its accuracy is superior to that of TBRG01 (refer to Table 2 in which the bucket sizes of TB01 and TB05 are 0.1 mm and 0.5 mm, respectively).

The first TB rain gauge was invented in 1662 by Christopher Wren and Robert Hooke. Since then, its improvement has primarily been focused on its accuracy, based on the development of both hardware and software. Concerning the hardware aspect, many studies have been performed that modify its aerodynamic shape to minimize the effects of wind, and working devices of TBs, including an extension to snowfall measurements (e.g., Das and Prakash 2011). The compensation of underestimation due to splash-out of rainfall water on tipping-bucket by comparing and calibrating the standard volume of water with the measured data has investigated (Shiraki et al. 2019).

In this study, we present an accuracy improvement of rainfall measurement by a new type of TB rain gauge: DTBG (Dual Tipping Bucket Gauge). It is composed of two tipping buckets (TBs): TB01 and TB05. The sizes of TB01 and TB05 are 0.1 mm and 0.5 mm, respectively. TB05 is introduced to catch splash-out of rainwater and/or overflow of rainwater from TB01. The remainder of this paper proceeds as follows. The observation site, specifications of used rain gauges, structure of DTBG, and methodology are described in Section 2. In Section 3, the data acquisition and analysis are explained. The inter-comparison of rainfall measurements by DTBG and Pluvio2 are also described in detail. The results of the current

Table 1 Comparisons of the characteristics of Pluvio2 and DTBG

Type of rain gauge	Pluvio2	DTBG (TB01/TB05)
Gauge characteristics	weighing sensor	tipping bucket
Receiver diameter	160 mm	200 mm
Rainfall measurement resolution	0.01 mm	0.1 mm/0.5 mm
Temporal resolution	6 s	1 min
Operating temperature	-50°C to 70°C	-20°C to 70°C
Manufacturer	OTT, Germany	GBM Inc., South Korea

study are given in Section 4. Finally, the summary and conclusions are presented in Section 5.

2 DTBG and Methodology

2.1 Used Rain Gauges

The data used in the present study were obtained from DTBG and Pluvio2, installed at the Cloud and Physics Observation Site (CPOS; 37.69 N, 128.76 E). The site is located at an altitude of 843 m (above sea level), Pyeongchang, South Korea. The distance between DTBG and Pluvio2 is approximately 25 m. The rainfall measurements were made by Pluvio2 with a wind shield. However, the measurements by DTBG were made without wind shield because it is difficult to secure an enough space for the wind shield. Most of considered rainfall events in this study has the low wind speed below 5 m/s except two cases (June 26 and July 2 in 2018). The wind-induced water loss was not counted in the rainfall data by DTBG. The characteristics of DTBG and Pluvio2 are given in Table 1, including

type, size, temporal resolution, accuracy, and manufacturer. The performance of DTBG was evaluated by using one-hour rainfall data from Pluvio2. The reason for this is that Pluvio2 (weighing rain gauge) is more accurate than TB rain gauges. Moreover, Pluvio2 was used here as a reference gauge because it was employed as one of the infield reference instruments for the WMO SPICE campaign (Colli et al. 2014). Pluvio2 also has higher sensitivity than TB rain gauges (Habib et al. 2001; Colli et al. 2013). The following is a more detailed description of Pluvio2.

The used Pluvio2 (version 400), a type of weighing gauge, was manufactured by OTT (2002), and has a bucket orifice opening of 400 cm² and a capacity of 750 mm. The Pluvio2 measures the weight of water content every 6 s with a measurement resolution of 0.01 mm. The rainfall intensity by Pluvio2 was used to assess the performance of DTBG, TB01, and TB05, based on an inter-comparison of rainfall amounts and rain intensities measured by DTBG, TB01, and TB05.

2.2 Shape and Structure of DTBG

DTBG is manufactured by GBM, and is designed to reduce systematic errors due to rainwater splash-out and/or overflow

Table 2 Comparison of daily accumulated rainfall by Pluvio2, DTBG, TB01, and TB05. Deviations between accumulated rainfall are given by Pluvio2-DTBG and TB05-TB01

Date (year/month/day)	Accumulated rainfall for 24 h (mm)				Deviation for accumulated rainfall (mm)	
	Pluvio2	TB01	TB05	DTBG	Pluvio2-DTBG	TB05-TB01
2018/04/24	29.3	26.4	29.5	29.5	-0.2	3.1
2018/05/02	23.5	22.9	22.5	22.9	0.6	-0.4
2018/05/06	19.9	19.2	19.0	19.2	0.7	-0.2
2018/05/08	22.9	21.3	21.0	21.3	1.6	-0.3
2018/05/17	55.1	52.6	54.5	54.5	0.6	1.9
2018/05/18	98.0	87.7	89.0	89.0	9.0	1.3
2018/06/10	27.9	26.2	26.5	26.5	1.4	0.3
2018/06/26	44.6	42.4	42.0	42.4	2.2	-0.4
2018/07/01	59.1	56.4	57.0	57.0	2.1	0.6
2018/07/02	40.1	37.8	38.0	38.0	2.1	0.2
2018/07/04	43.0	39.8	40.0	40.0	3.0	0.2
2018/07/05	34.0	29.0	29.0	29.0	5.0	0.0
2018/07/06	99.2	88.2	88.0	88.2	11.0	-0.2
2018/07/09	22.1	20.7	21.0	21.0	1.1	0.3



from TB01. Its shape and structure are given in Fig. 2. DTBG is composed of five main components: A (heater); B (funnel of 0.1 mm); C (TB of 0.1 mm); D (funnel of 0.5 mm); and E (TB of 0.5 mm). Part A in the right panel represents a heater with a thermistor. The heater is used to melt snow accumulated in the orifice during wintertime. Part B is the primary receiver, which collects rainfall, and then drains rain water into part C. Part C is TB01 (upper bucket), corresponding to the primary TB with a measurement resolution of 0.1 mm, and functions like a single TB rain gauge. Filling TB01 with water of nominal volume, the weight of the water allows tipping of the bucket, dumping the water into part D, which constitutes a secondary funnel. The dumped water through D flows into part E, TB05 (lower bucket) with a measurement resolution of 0.5 mm. TB05 works in the same manner as TB01. The temporal resolution of DTBG is set one minute as given in Table 2 (minimum is 10 second).

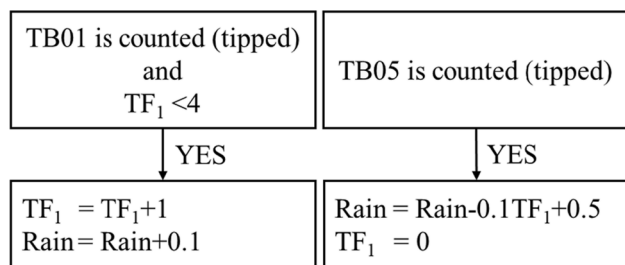


Fig. 1 The algorithm flowchart of the DTBG operation

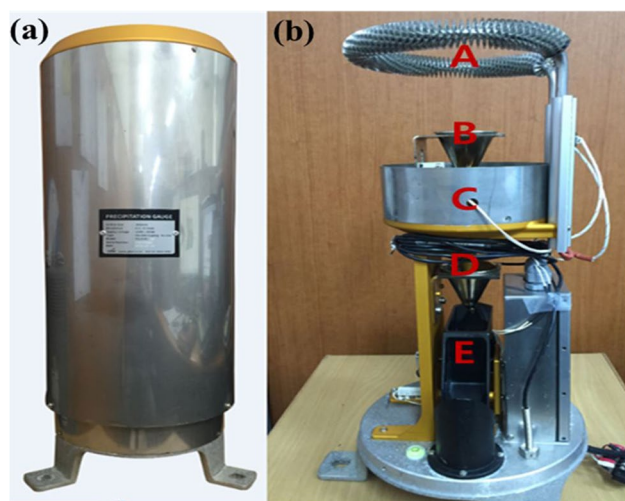


Fig. 2 Shape and major parts of DTBG: (a) exterior cylinder; and (b) internal structure. DTBG is composed of five main components: A (heater); B (funnel of 0.1 mm); C (bucket of 0.1 mm); D (funnel of 0.5 mm); and E (bucket of 0.5 mm)

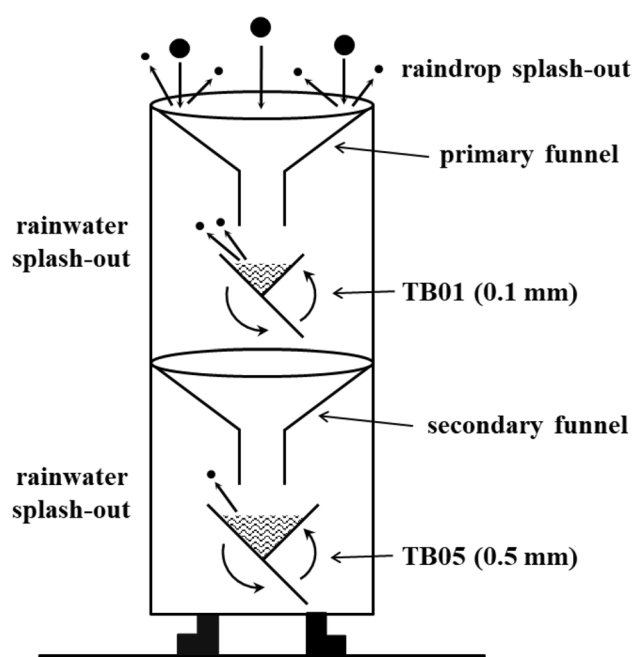


Fig. 3 Two types of splash-out by DTBG: raindrop splash-out; and rainwater splash-out

2.3 Operation Principles of DTBG

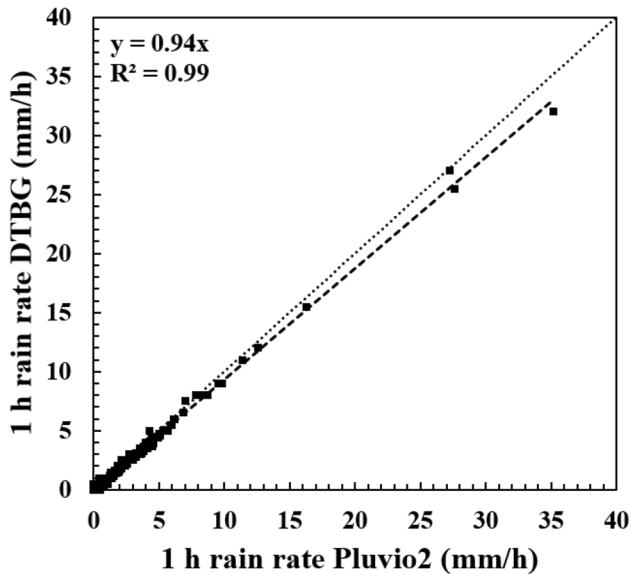
A conceptual illustration of how DTBG works and improves the accuracy of rain rate is given in Figs. 1, 2 and 3. The TB01 is connected to the secondary funnel. The channel enables catching of the rainwater splash-out and/or overflow from TB01. The channel is connected to TB05 (lower bucket). Then, TB05 measures rain rate like a single TB rain gauge, based on the tipping frequency of the bucket. The catchment of splash-out and/or overflow enhances rain intensity, leading to accurate rainfall measurement. The rain rainwater splash-out and/or overflow from TB05, however does not contribute at all to catchment enhancement because it is drained away from DTBG.

Table 3 Statistical comparison for three pairs of rain gauges with the events shown in Table 2

Rainfall statistics	Bias(mm)	Absolute bias(mm)	Fractional bias (%)	Fractional absolute bias (%)	R ²
Pluvio2 vs. TB01	1.851	1.864	0.084	0.085	0.99
Pluvio2 vs. TB05	1.657	1.708	0.075	0.077	0.99
Pluvio2 vs. DTBG	1.578	1.631	0.071	0.074	0.99

Table 4 Accuracy comparison for three pairs of rain gauges with one-hour rain rate

Rainfall statistics	Bias (mm/h)	RMSE (mm/h)	R ²
Pluvio2 vs. TB01	0.142	0.358	0.99
Pluvio2 vs. TB05	0.127	0.350	0.99
Pluvio2 vs. DTBG	0.124	0.325	0.99

**Fig. 4** Scatter plot of hourly rain rates by Pluvio2 and DTBG. The plot is made by using the data of all rainfall events shown in Table 2

Concerning the measurements of rainfall by DTBG, the rain rates from TB01 and TB05 are determined by their tipping frequencies (TF). Then, the higher rain rate from either TB01 or TB05 is taken as the rate by DTBG. This constitutes a major advantage of DTBG over a single TB rain gauge. The following conceptual and simplified algorithm for the DTBG operation are applied to hourly data:

If $[0.5 \text{ mm} \times \text{TF}_5(\text{h}^{-1}) > 0.1 \text{ mm} \times \text{TF}_1(\text{h}^{-1})]$

then rain rate (mm/h) = $0.5 \text{ mm} \times \text{TF}_5(\text{h}^{-1})$,

else rain rate (mm/h) = $0.1 \text{ mm} \times \text{TF}_1(\text{h}^{-1})$, (1)

where TF_1 and TF_5 represent the tipping frequency of TB01 and TB05, respectively. Figure 1 shows the actual operation method from the observed counts of two tipping bucket to record the rainfall measurement data. As shown in Fig. 1, the measurement priority of TBs is given to TB05 and TB01 has only the supplementary role to record the accumulated rainfall.

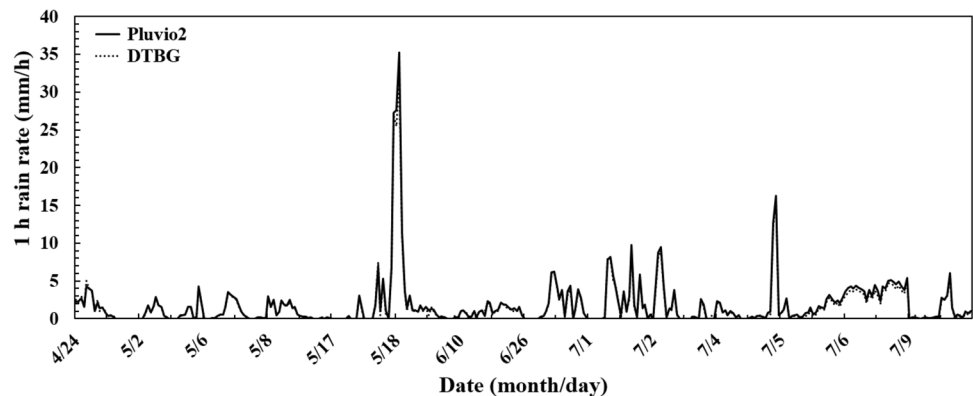
3 Statistics for DTBG Evaluation

The performance of DTBG was evaluated by using consistency and discrepancy (Guo and Yuanbo 2016). Consistency refers to the proximity or similarity between two data sets. Here, consistency is assessed by using a coefficient of determination (R^2). The coefficient is a statistical measure of how well a regression line approximates the points of two data sets (a_i, b_i). For the present study, a_i and b_i represent rainfall rates by Pluvio2 and DTBG, respectively. Generally, a higher coefficient indicates a better fit for the two data sets. The following equation is used to calculate the coefficients of determination (R^2):

$$R^2 = \frac{\sum_{i=1}^n \{(a_i - \bar{a})(b_i - \bar{b})\}^2}{\sum_{i=1}^n (a_i - \bar{a})^2 \times \sum_{i=1}^n (b_i - \bar{b})^2}, \quad (2)$$

where $\bar{a}$ and $\bar{b}$ represent the mean values of a_i and b_i , respectively; and n is the number of data pairs.

The discrepancy between the reference and estimated quantities is specified by using bias, absolute bias, fractional bias, and absolute fractional bias. These biases are used to assess overall deviation from the reference, and are calculated by applying the following equations for the rainfall data (Tokay et al. 2013):

Fig. 5 Time series of hourly rain rates by DTBG and Pluvio2. The plot is made for all rainfall events given in Table 2

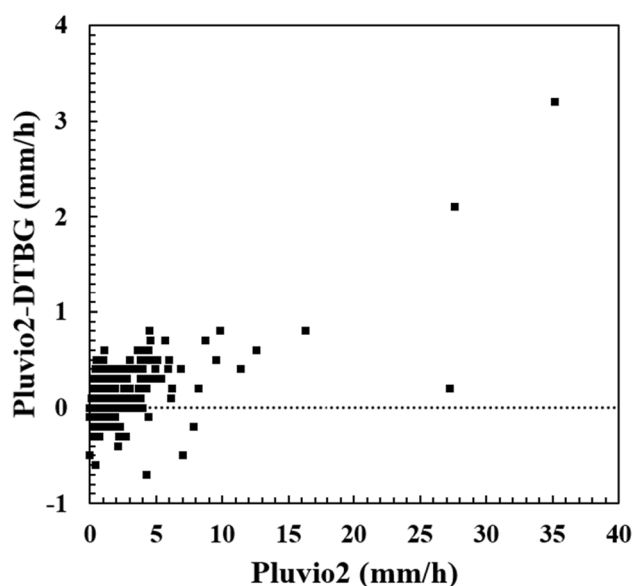


Fig. 6 Scatter plot of Pluvio2-DTBG (mm/h) against rain rate from Pluvio2

$$\text{bias} = \frac{1}{n} \sum_{i=1}^n (a_i - b_i), \quad (3)$$

$$\text{absolute bias} = \frac{1}{n} \sum_{i=1}^n |a_i - b_i|. \quad (4)$$

The fractional bias and absolute bias between the data of two rain gauges (a , b) are also calculated by using the following equations:

$$\text{fractional bias}(\%) = \frac{\text{bias}}{[a, b]}, \quad (5)$$

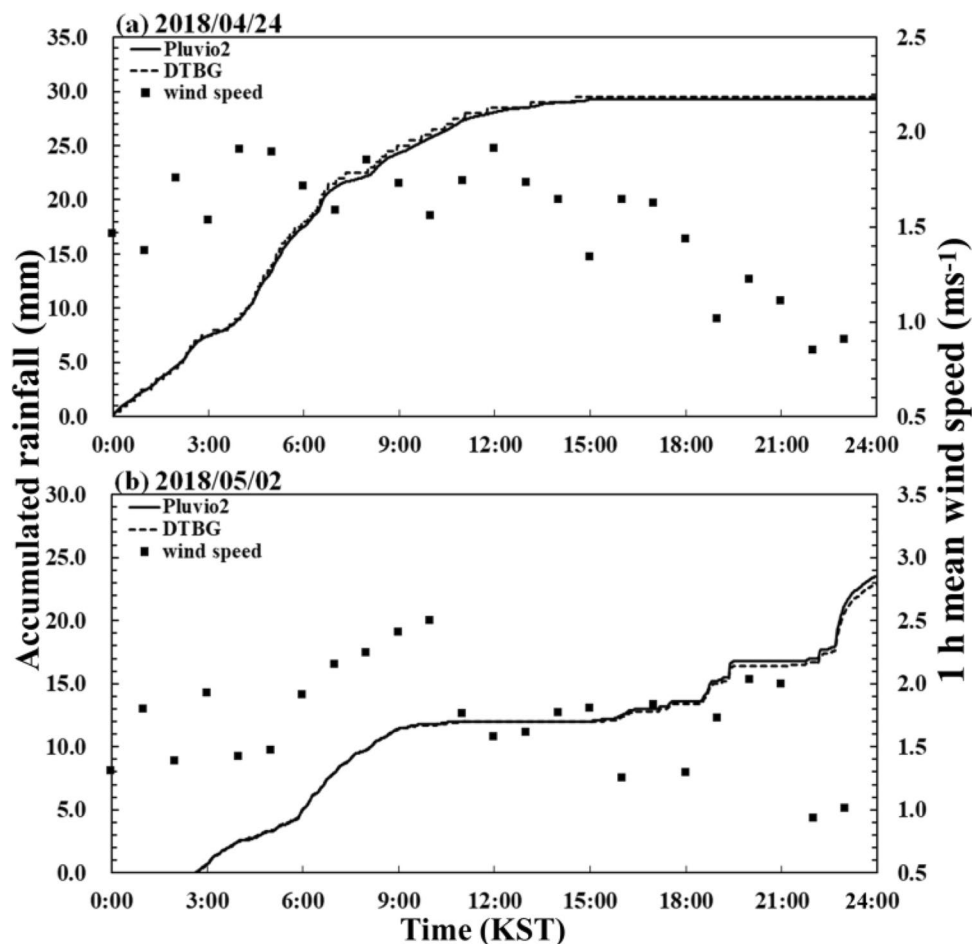
$$\text{fractional absolute bias}(\%) = \frac{\text{absolute bias}}{[a, b]}, \quad (6)$$

where $[a, b]$ is the mean value between the data of two rain gauges (a , b), and is expressed as follows:

$$[a, b] = \frac{1}{n} \sum_{i=1}^n \frac{(a_i + b_i)}{2}. \quad (7)$$

To assess how well the rain rate of DTBG matches the rate of Pluvio2, RMSE(Root Mean Square Error) is

Fig. 7 Accumulated rainfall with time, including 1 h mean wind speed for rainfall events 1 and 2 in Table 2



calculated by applying Eq. (8) to the rain rate data from DTBG and Pluvio2:

$$\text{RMSE} = \left[\frac{1}{n} \sum_{i=1}^n (a_i - b_i)^2 \right]^{\frac{1}{2}}. \quad (8)$$

RMSE here is a metric that tells us the average difference between the rain rates of DTBG and the rates of Pluvio2 (reference gauge).

4 Data and Analysis Results

4.1 Data

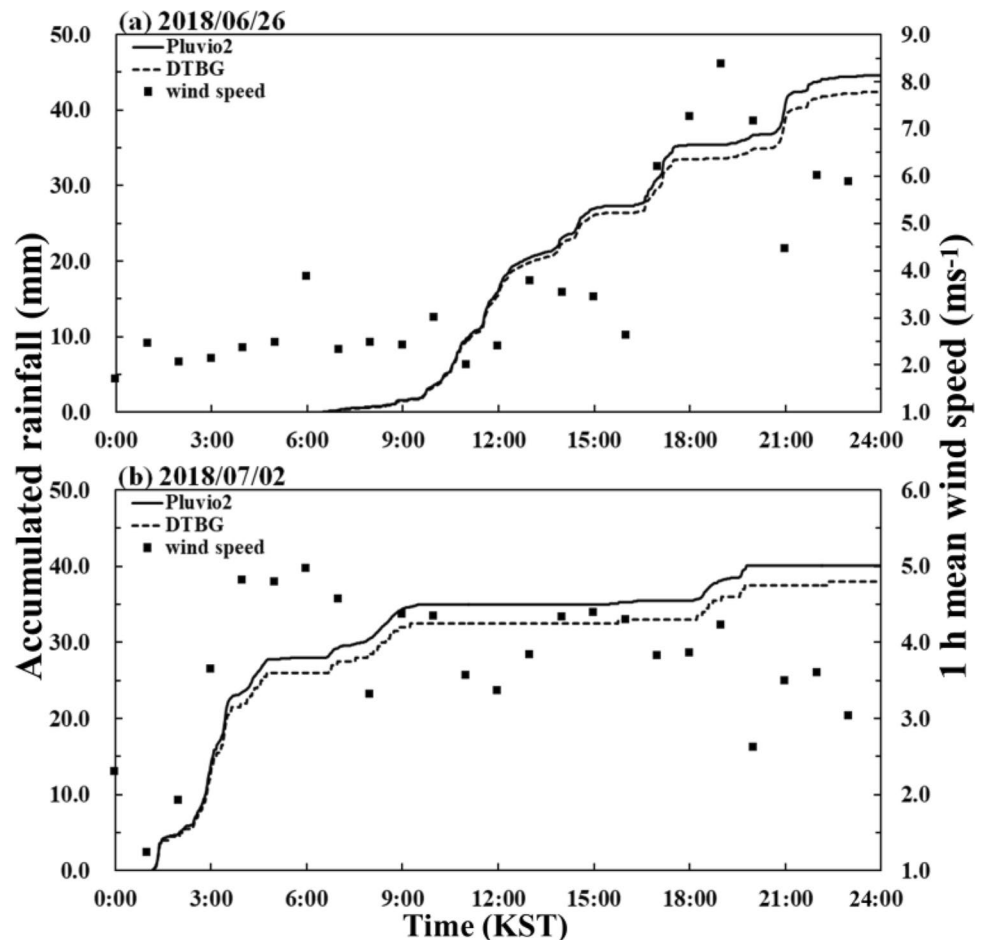
The data used for the study is rainfall measurements made by DTBG and Pluvio2, and wind data from an automatic weather station at CPOS site. The data is obtained for 14 rainfall events during the period from 24th April to 9th July, 2018. The rainfall data is used for accuracy evaluation of DTBG against Pluvio2. The wind data is used to analyze effect of wind on rainfall observation of the two instruments during the observation period. The temporal resolution of DTBG and Pluvio2 are one minute and 0.6 s, respectively.

The data from DTBG and Pluvio2 is converted into one-hour rain rate for their comparison.

4.2 Data Analysis and Results

The present analysis is mainly focused on two aspects of DTBG: (1) the accuracy of DTBG against Pluvio2; and (2) the mechanical performance of DTBG with rain rates. The accuracy of DTBG is evaluated by using Tables 2, 3 and 4. Table 2 presents the daily rainfall amounts for 14 rainfall events measured by Pluvio2, TB01, TB05, and DTBG, respectively. In Table 2, there are two deviations, i.e., Pluvio2-DTBG and TB05-TB01, the former of which represents the difference between the daily rainfall amount by Pluvio2 and DTBG. According to the table, the values of the former deviations are all positive, except for event 1. This suggests that DTBG somewhat undercatches rainfall compared to rainfall by Pluvio2 (reference gauge). The latter deviation indicates the difference between the daily rainfall amount by TB05 and TB01. For most of the events, as shown in the table, the deviations between TB05 and TB01 are less than the ones between Pluvio2 and DTBG. This is because the mechanical types of TB05 and TB01 are identical; whereas, the types of Pluvio2 and DTBG are markedly different from each other.

Fig. 8 Accumulated rainfall with time, including 1 h mean wind speed for rainfall events 8 and 10 in Table 2

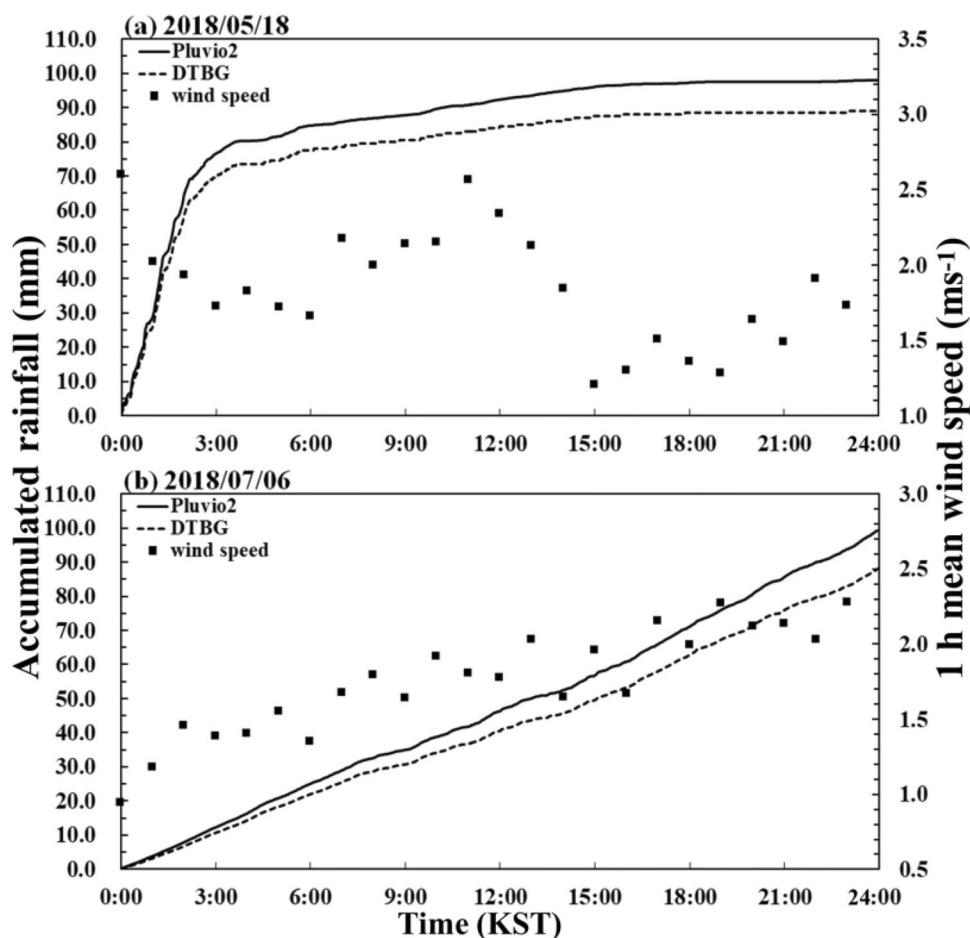


The accuracy of DTBG was assessed, based on Table 3. In the table, four different biases and three coefficients of determination (R^2) are given for three pairs of rain gauges. All statistics in Table 3 are obtained by applying the equations in Section 2.3 to all rainfall events in Table 2. As shown in Table 3, all biases of Pluvio2 vs. DTBG have the lowest values compared to those of Pluvio2 vs. TB01 and Pluvio2 vs. TB05. The value of R^2 for Pluvio2 vs. DTBG is the same as those for Pluvio2 vs. TB01 and Pluvio2 vs. TB05. Moreover, the overall statistics of Pluvio2 vs. DTBG indicate that the rainfall data by DTBG are much closer to those by Pluvio2 rather than those by a single TB rain gauge, i.e., TB01 or TB05. The analysis indicates that DTBG achieves higher accuracy than a single TB rain gauge. Figure 4 is a scatter plot of rain rates by Pluvio2 and DTBG. The figure was plotted by using 1 h rainfall rates for all rainfall events measured by Pluvio2 and DTBG. The value of R^2 between the two rain rates is 0.99, indicating a very high correlation between rainfall measurements by Pluvio2 and DTBG. The red dotted-line in Fig. 4 is the regression line, showing that the value of R^2 is 0.99. The dotted-line in Fig. 4 is the regression line of $y = 0.94x$, indicating that the reduction of 6% on the rain rates, compared to those by Pluvio2.

Figure 5 shows a plot of time series of hourly rain rates by DTBG and Pluvio2, based on all rainfall events given in Table 2. The figure shows the rainfall intensities by DTBG are fairly consistent with those by Pluvio2. Concerning Table 4, there are three statistical parameters: bias, RMSE and R^2 . The parameters are calculated from one-hour rain rate. As shown in Table 4, the values of three parameters are the smallest for Pluvio2 vs. DTBG. This indicates that the rain rates by DTBG are much closer to those by Pluvio2 rather than those by a single TB rain gauge, i.e., TB01 or TB05. To analyze the performance of DTBG with rain rates, Fig. 6 is plotted by using one-hour rain rates from Pluvio2 and the deviation of rain rates between pluvio2 and DTBG. For the rain rate less than about 16 mm/h, the difference between Pluvio2 and DTBG is smaller than ± 1 mm. This indicates the use of tipping bucket rain gauge needs the caution for the strong rainfall larger than 16 mm/h.

Depending on daily rainfall (R_{day}), all events in Table 2 are divided into three groups: (1) $R_{\text{day}} < 30$ mm; (2) $30 \text{ mm} \leq R_{\text{day}} \leq 50$ mm; and (3) $R_{\text{day}} > 50$ mm. The groups are specified to analyze the performance of DTBG. Concerning the group with $R_{\text{day}} < 30$ mm, events 1 and 2 are selected from Table 2. For these events, the temporal variations of accumulated rainfall and 1 h mean wind speed are given

Fig. 9 Accumulated rainfall with time, including 1 h mean wind speed for rainfall events 6 and 13 in Table 2



in Fig. 7. The figure shows that the temporal variations of accumulated rainfall by DTBG and Pluvio2 are almost the same. This indicates that there is no significant undercatchment of rainfall by DTBG. Regarding wind effect, the average wind speeds of event 1 and 2 are 1.5 ms^{-1} and 1.7 ms^{-1} , respectively. These speeds are markedly less than the criterion of wind speed (approximately 5 ms^{-1} to 6 ms^{-1}) on rainfall undercatchment established by Duchon and Biddle (2010). It is shown in Fig. 8 that in the observational difference between DTBG and Pluvio2 becomes large with the higher wind speed than 5 m/s , which agrees with the results of Duchon and Biddle. However, the relationship between the wind speed and error of TBs needs more investigations.

For the second group ($30 \text{ mm} \leq R_{\text{day}} \leq 50 \text{ mm}$), event 8 and 10 are selected from Table 2. Figure 8 concerns rainfall events 8 and 10 in Table 2. As shown in the figure, the temporal variations of accumulated rainfall by DTBG and Pluvio2 are almost the same up to the accumulated rainfall of approximately 20 mm. After passing the rainfall level of approximately 20 mm, the difference between rainfall by DTBG and Pluvio2 increases with time, showing an increase of undercatchment by DTBG with time. This behavior is dissimilar to that presented in Fig. 7. The average wind speeds of events 8 and 10 are 3.8 ms^{-1} and 3.7 ms^{-1} , respectively. Since the speeds are less than the criterion of wind speed (5 ms^{-1} to 6 ms^{-1}) on rainfall undercatchment (Duchon and Biddle 2010), the wind effect on the undercatchment of rainfall by DTBG is deemed to be insignificant. Regarding the third group ($R_{\text{day}} > 50 \text{ mm}$), event 6 and 13 are selected from Table 2.

The daily rainfall is approximately 100 mm for the two events, corresponding to the cases of heavy rainfall. According to Fig. 9a for event 6, the rainfall by DTBG and Pluvio2 are almost the same up to the accumulated rainfall of approximately 45 mm. After passing this rainfall level, the difference between the rainfall by DTBG and Pluvio2 steadily increases with time. In contrast with Figs. 9a and b for event 13 shows that the difference between rainfall by DTBG and Pluvio2 steadily increases with time from the beginning of the rainfall. The average wind speed of events 6 and 13 are both 1.8 ms^{-1} . Therefore, the wind effect on the undercatchment of rainfall for the two rainfall events is regarded as insignificant. From the analysis of Figs. 7, 8 and 9, it can be concluded that the tipping processes of DTBG depends highly on the rainfall events, except when the daily rainfall is less than 30 mm.

5 Summary and Conclusions

The performance of DTBG was assessed based on rainfall data obtained for 14 rainfall events (24th April to 9th July, 2018). Our data analysis indicates that DTBG obtains higher accuracy in rainfall measurements, compared to a single TB

rain gauge. The higher accuracy of DTBG is a consequence of the combination of TB01 and TB05. The result was demonstrated by inter-comparisons of accumulated rainfall amounts for all rainfall events by Pluvio2 and DTBG. The mean fractional bias of DTBG against Pluvio2 is 0.07%, indicating that the accuracy of DTBG is comparable to that of Pluvio2. Concerning the cases of heavy rainfall, however, a significant difference exists between rainfall amounts by DTBG and Pluvio2. The fractional bias of DTBG against Pluvio2 is approximately 10% for the cases of heavy rainfall. In conclusion, this study has shown that for rainfall measurements, DTBG achieves substantially higher performance than a singular TB rain gauge.

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Chang, Ki-Ho: research proposal, funding for the study, corresponding author;

Kim, Kyung-Eak: organization of the research paper and comments on the manuscript;

Bang, Ki Seok: company CEO of development and maintenance of DTBG.

Data Availability The data used are available upon request (Dr. Ki-Ho Chang: khchang@korea.kr).

Declarations

Conflict of Interest Not applicable.

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